Biochemistry



Section M

Nitrogen metabolism

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Section M

Nitrogen metabolism

- 1. Reduction of N_2 into ammonia (NH_3 or NH_4^+)
- 2. Synthesis of the 20 amino acids.
- 3. Amino acid degradation



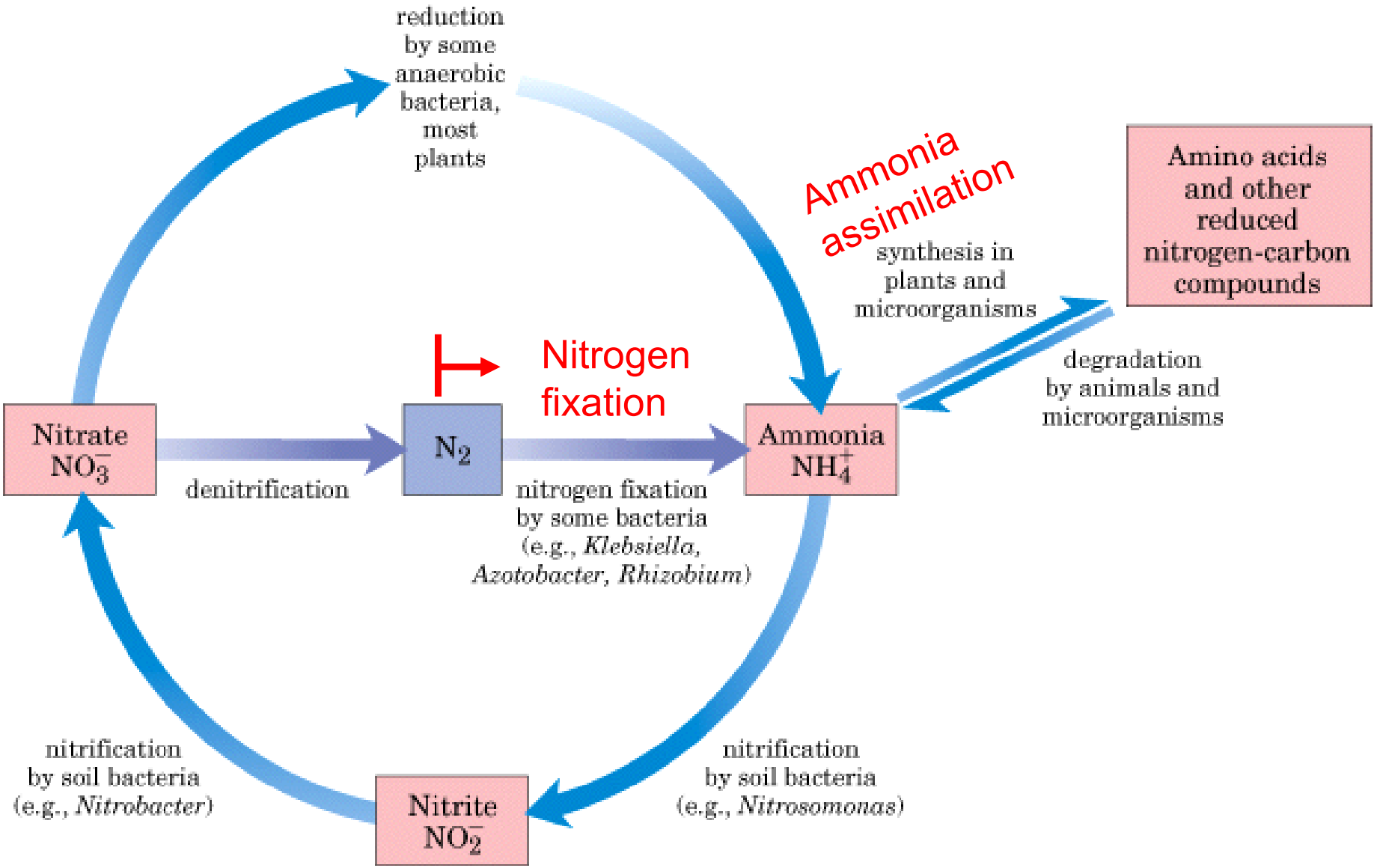
M1 Nitrogen fixation and assimilation

The nitrogen cycle

Nitrogen fixation

Nitrogen assimilation




reduction by some anaerobic bacteria, most plants
Ammonia assimilation
synthesis in plants and microorganisms
Amino acids and other reduced nitrogen-carbon compounds
degradation by animals and microorganisms
Nitrogen fixation
Nitrate NO_3^-
denitrification
N_2
nitrogen fixation by some bacteria (e.g., *Klebsiella, Azotobacter, Rhizobium*)
Ammonia NH_4^+
nitrification by soil bacteria (e.g., *Nitrobacter*)
Nitrite NO_2^-
nitrification by soil bacteria (e.g., *Nitrosomonas*)

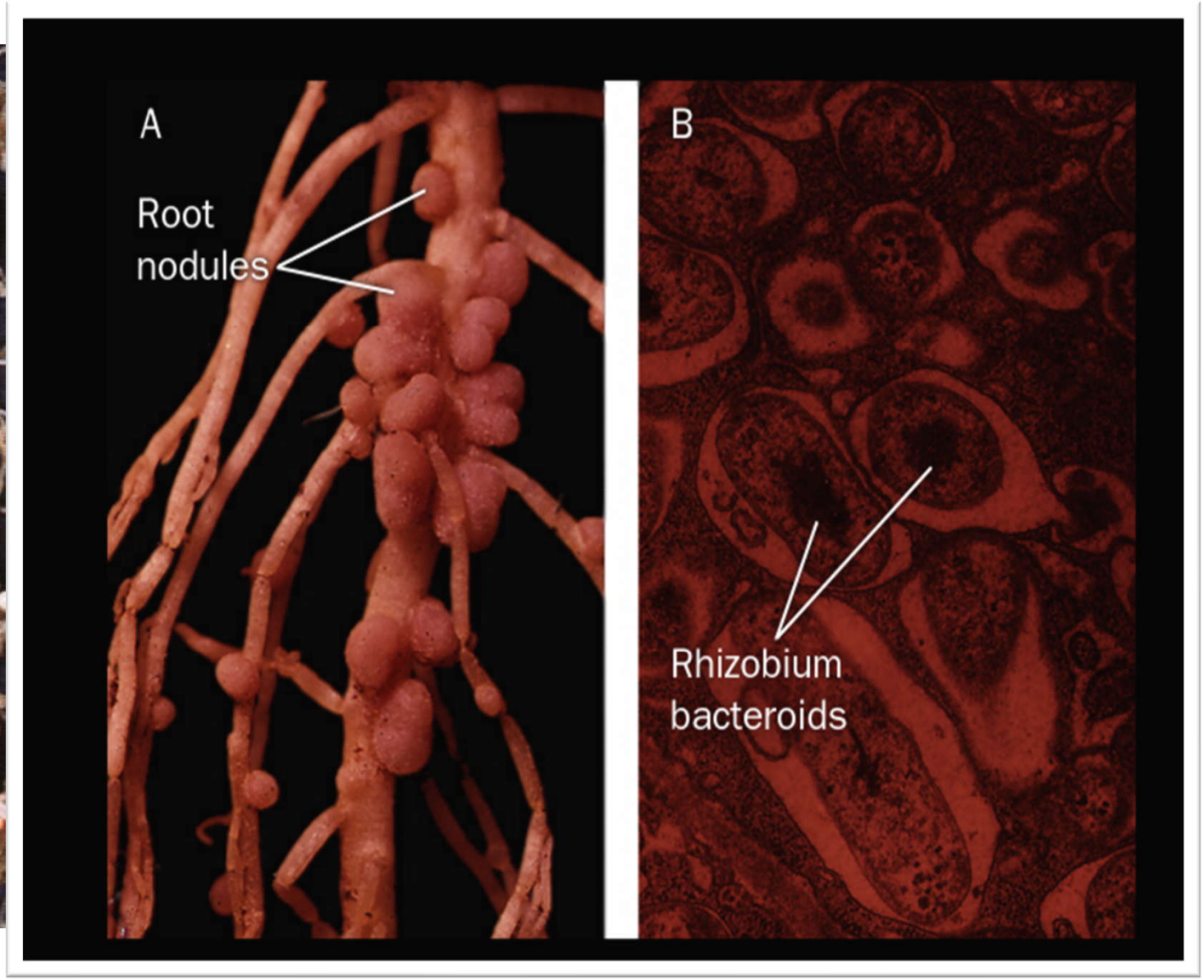
Nitrogen fixation

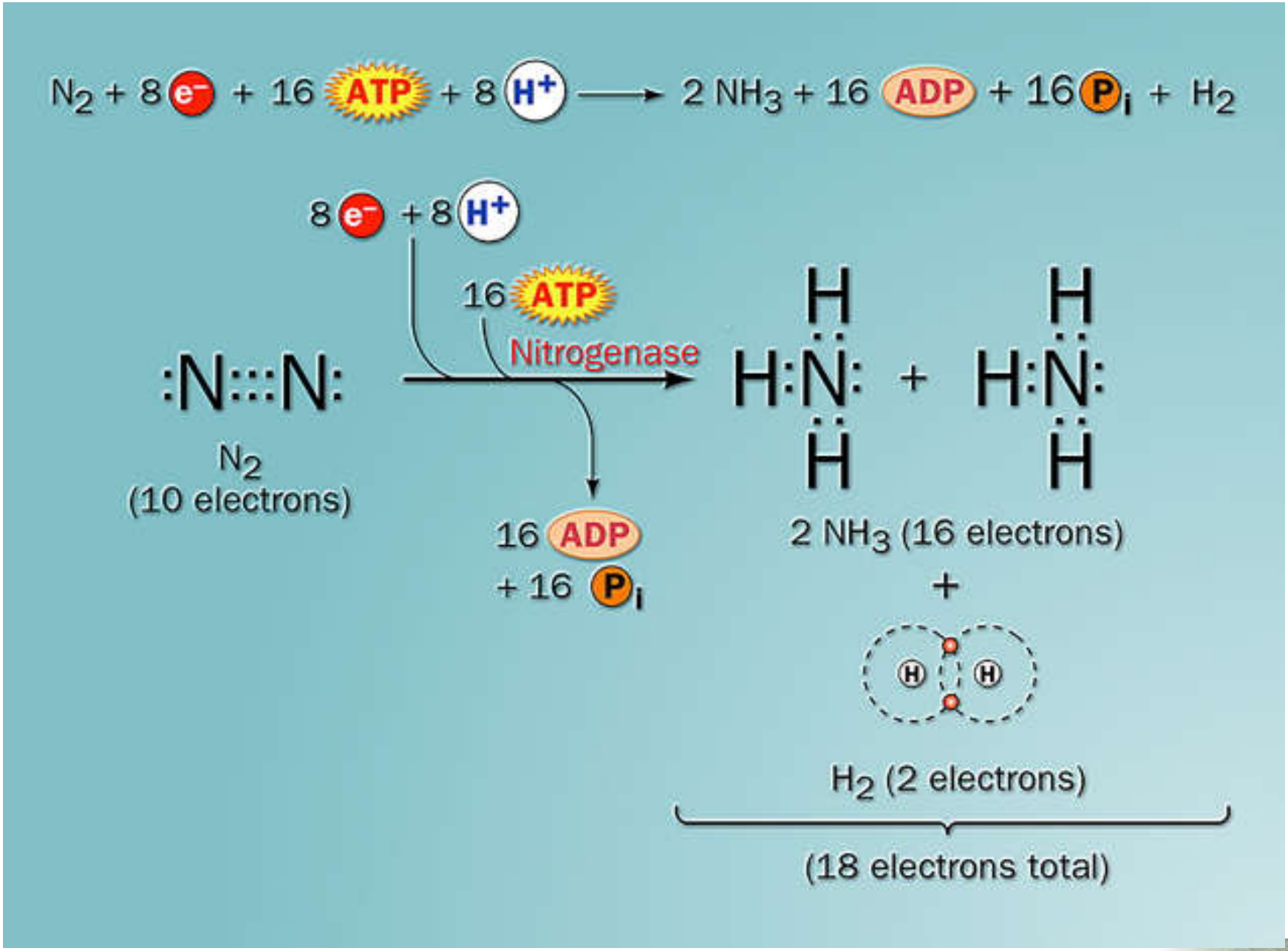
- The nitrogen in amino acids, purines, pyrimidines and other biomolecules ultimately comes from atmospheric nitrogen.
- Cyanobacteria (蓝藻细菌, photosynthetic) and rhizobia (根瘤菌, symbiont) can fix N_2 into NH_3.
- The reduction of N_2 to NH_3 is thermodynamically favorable :
-
-



Certain bacteria can fix N_2 into ammonia






N2 + 8 e− + 16 ATP + 8 H+ → 2 NH3 + 16 ADP + 16 Pi + H2
8 e− + 8 H+
16 ATP
Nitrogenase
:N:::N:
N2
(10 electrons)
16 ADP
+ 16 Pi
2 NH3 (16 electrons)
+
H2 (2 electrons)
(18 electrons total)

Nitrogen fixation

The nitrogenase complex in certain bacteria catalyzes the conversion of N_2 to NH_3, which is the ultimate source of nitrogen for all the nitrogen-containing biomolecules.



Electrons are transferred through a series of carriers to N_2 for its reduction on the nitrogenase complex

$$N_2 + 8H+ + 8e- + 16ATP + 16H_2O$$

$$\rightarrow 2NH_3 + H_2 + 16ADP + 16P_i$$



M2 Amino Acid Metabolism

- Biosynthesis of amino acids
- Amino acid degradation
- Transamination
- Oxidative deamination of glutamate
- Amino acid oxidation
- Metabolism of phenylalamine

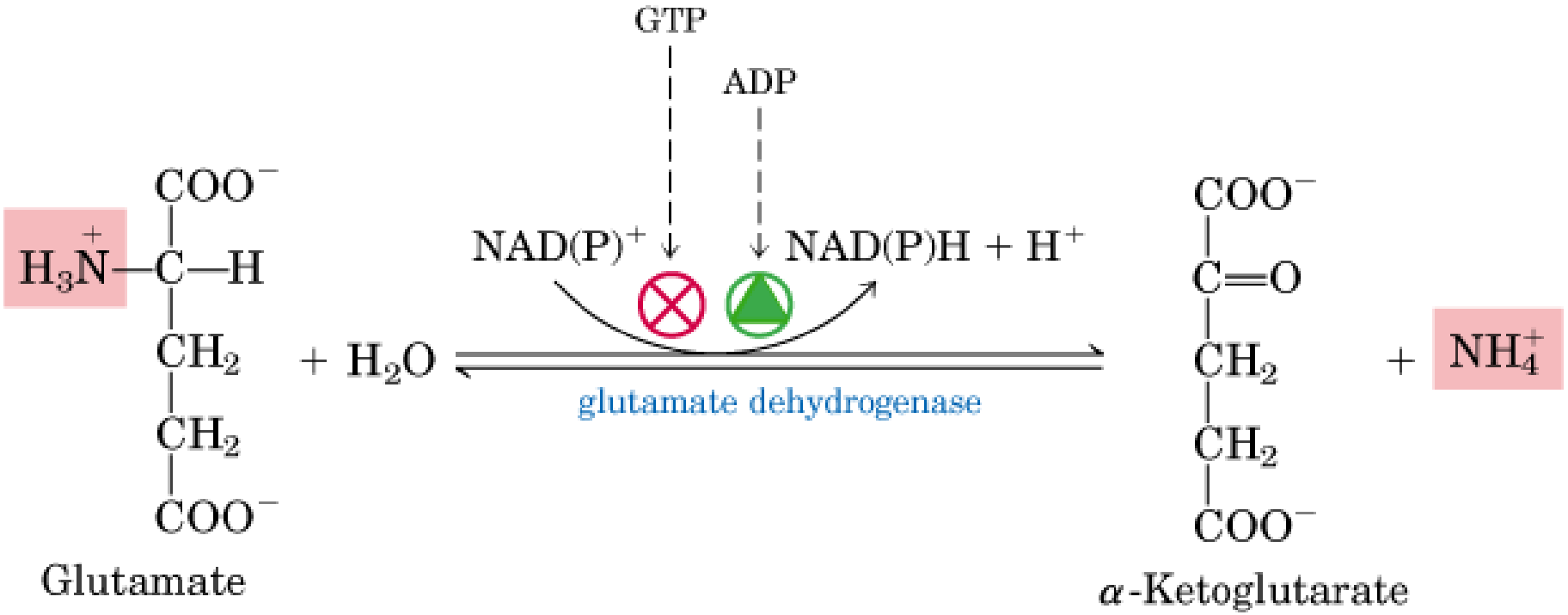


Ammonia assimilation

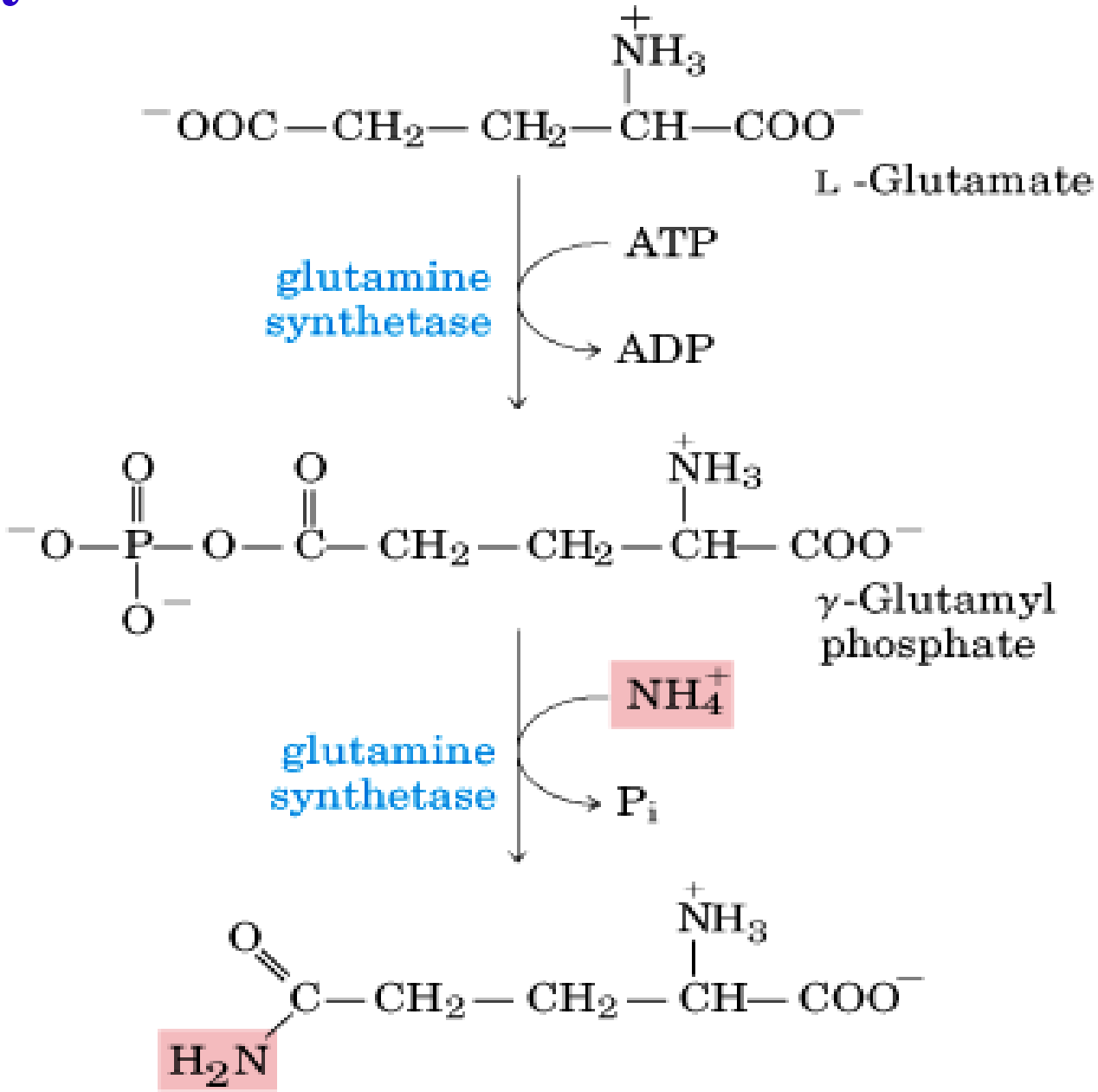
Ammonia is incorporated into biomolecules through Glutamate and glutamine.

Reduced nitrogen in the form of NH_4^+ is assimilated into organic nitrogen-containing compounds by the action of glutamate dehydrogenase and glutamine synthetase .

glutamate dehydrogenase



glutamine synthetase



glutamine synthetase

Glutamate + NH_3 + ATP ⟶ Glutamine + ADP + Pi

Glutamine + Ketoglutarate ⟶ (Glutamate synthase; **NADPH**, **$NADP^+$**) 2Glutamate

形成Gln既是氨同化的一种方式，又可消除过高氨浓度带来的毒害，还可以作为氨的供体，用于Glu的合成。



Biosynthesis of amino acids

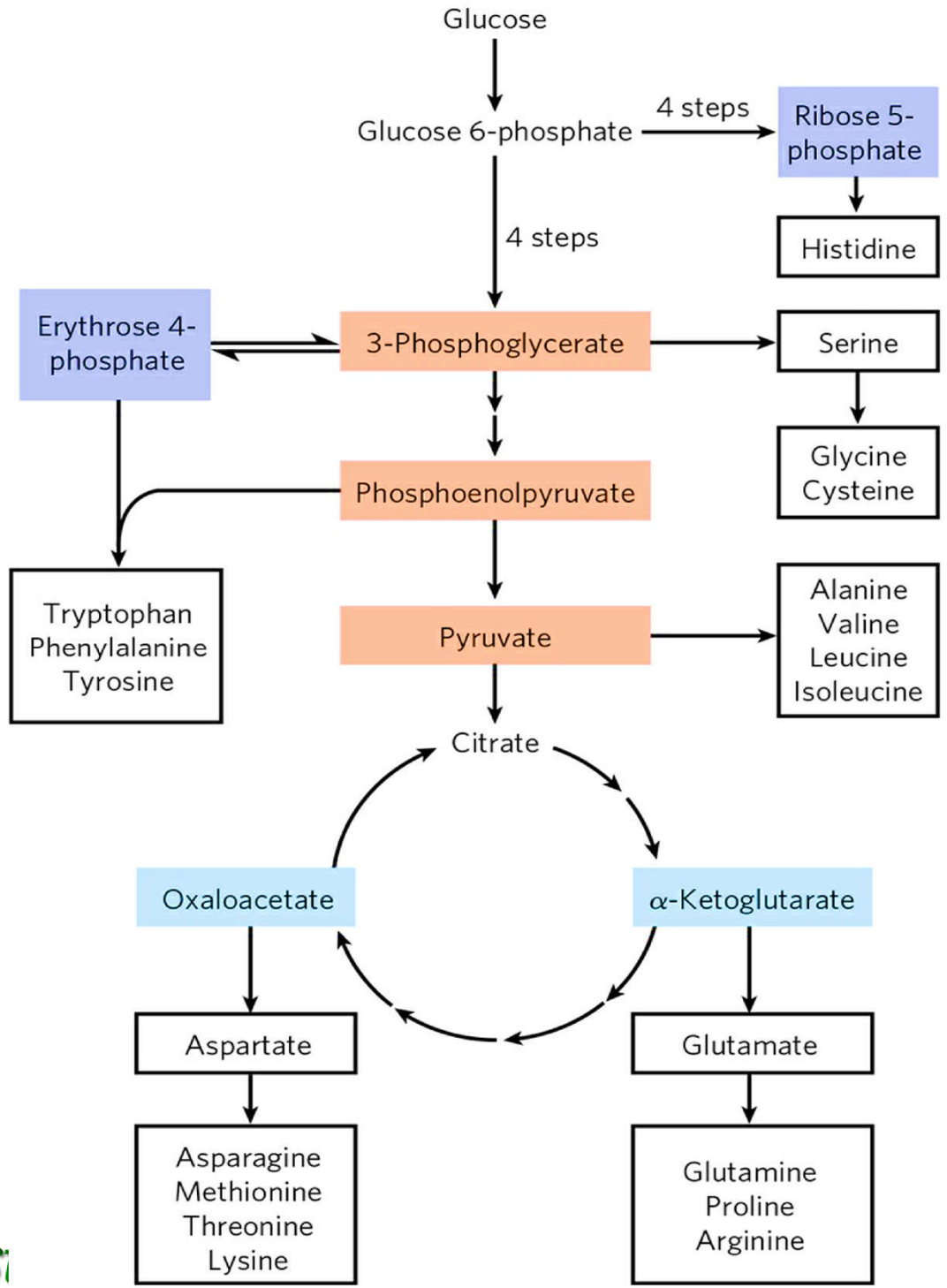
The amino acids can be grouped into six biosynthetic families depending on the metabolic intermediate from which their carbon skeleton is derived.

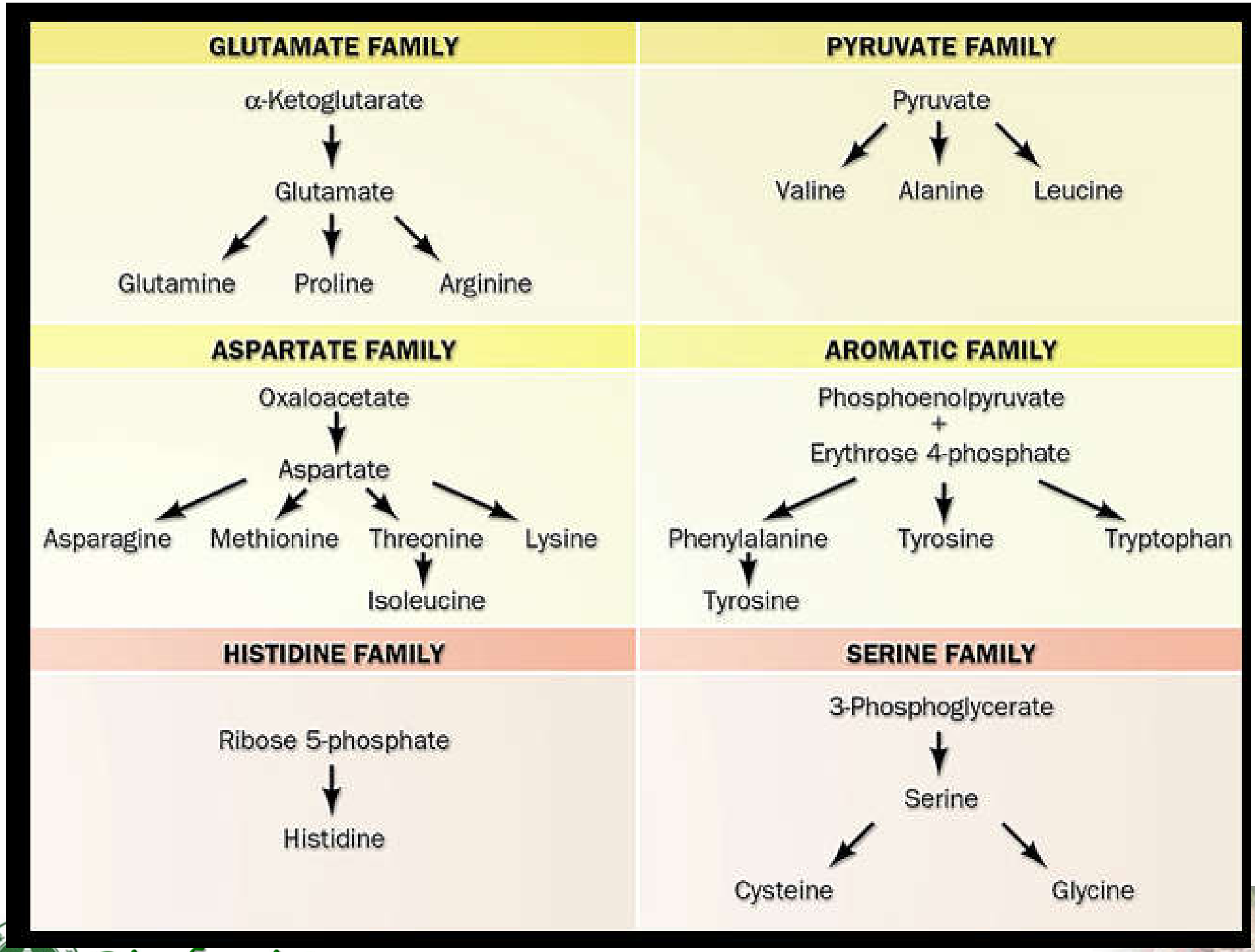


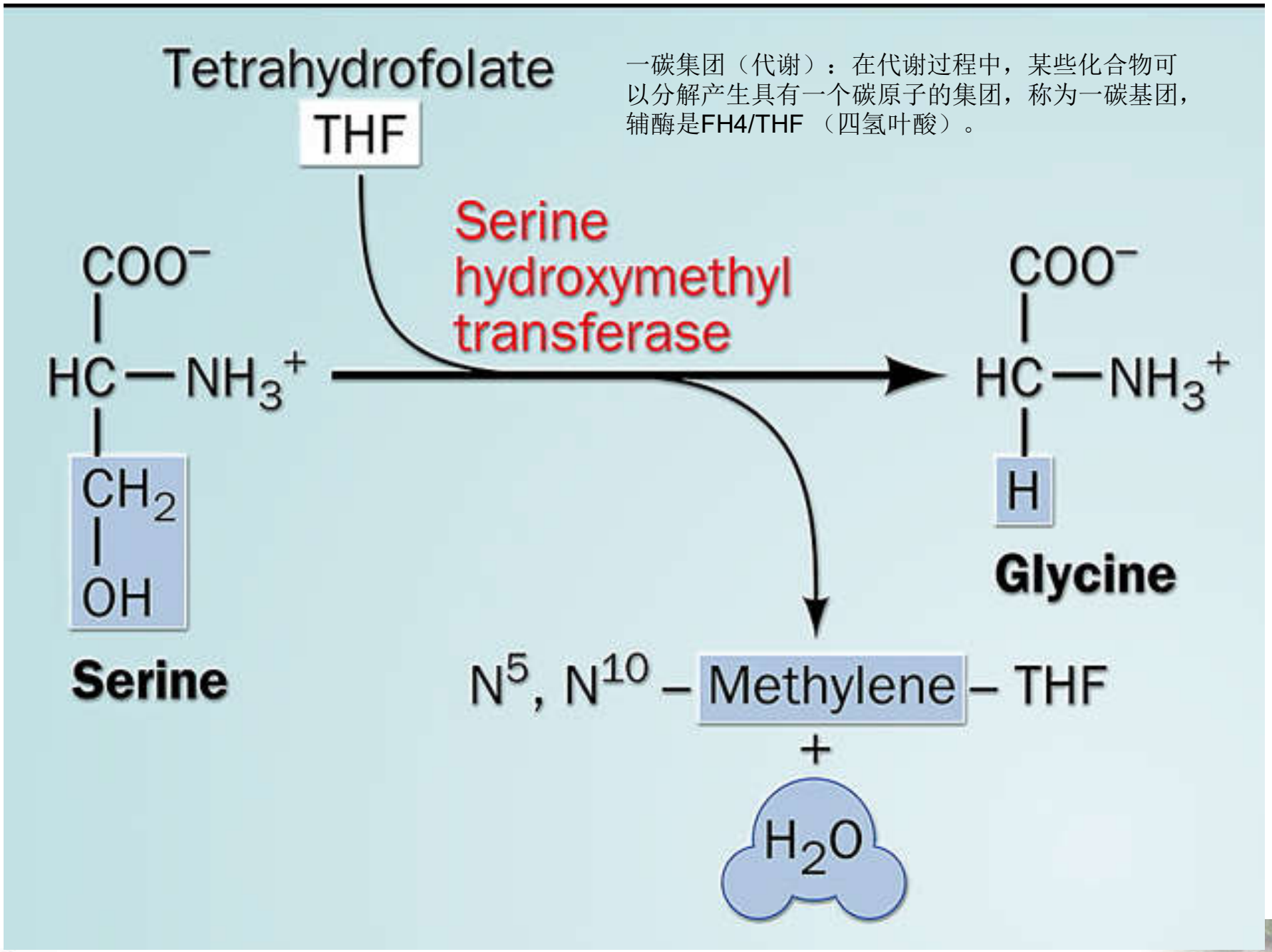
The 20 amino acids are synthesized mainly from intermediates of glycolysis, citric acid cycle, or pentose phosphate pathway in bacteria and plants.

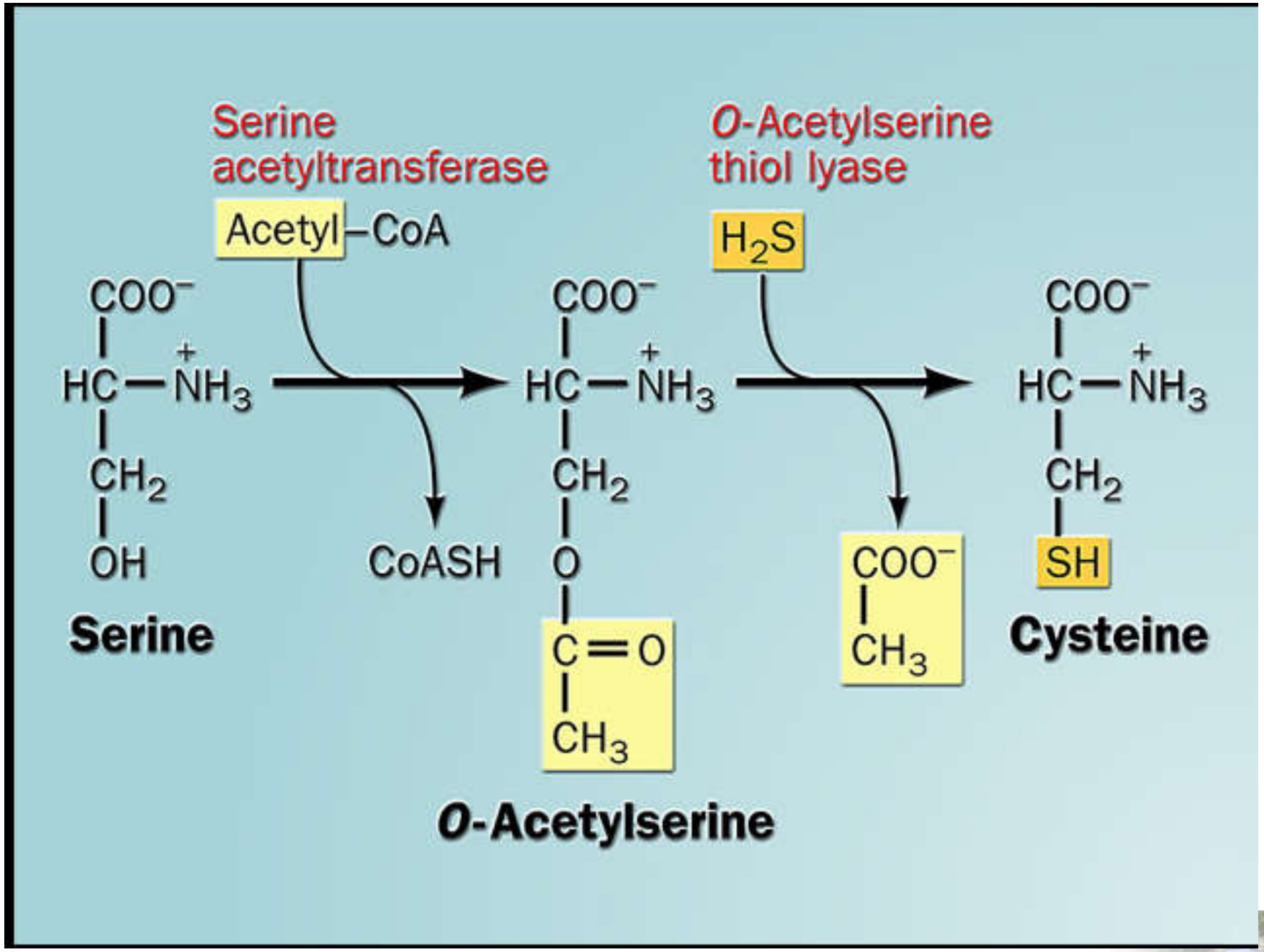


aromatic

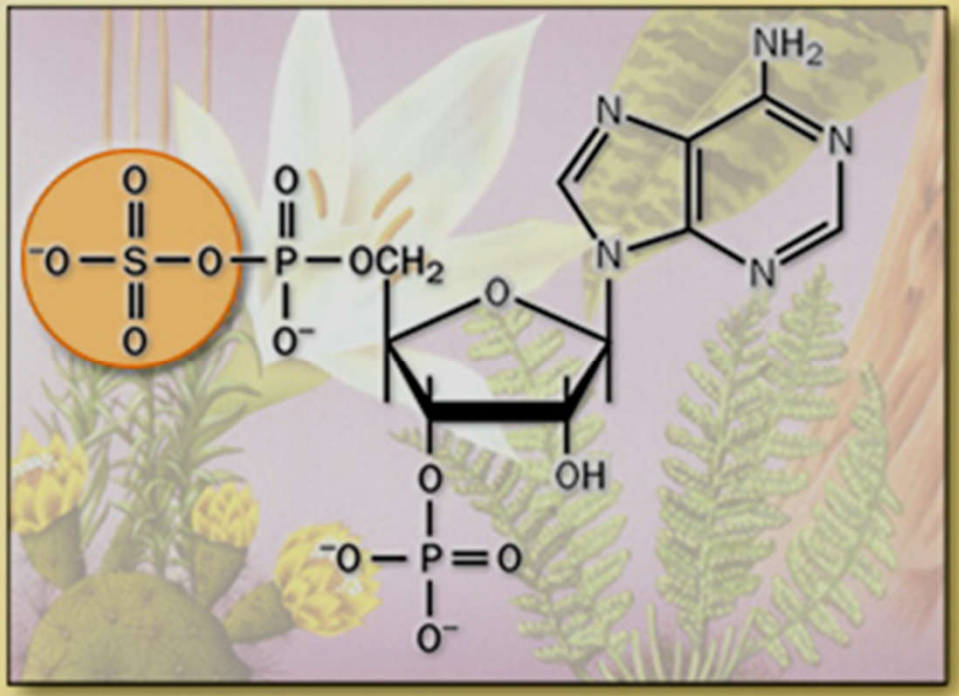

Glucose
Glucose 6-phosphate
4 steps
Ribose 5-phosphate
Histidine
4 steps
Erythrose 4-phosphate
3-Phosphoglycerate
Serine
Glycine
Cysteine
Phosphoenolpyruvate
Tryptophan
Phenylalanine
Tyrosine
Pyruvate
Alanine
Valine
Leucine
Isoleucine
Citrate
Oxaloacetate
α-Ketoglutarate
Aspartate
Glutamate
Asparagine
Methionine
Threonine
Lysine
Glutamine
Proline
Arginine


GLUTAMATE FAMILY
α-Ketoglutarate
Glutamate
Glutamine
Proline
Arginine
PYRUVATE FAMILY
Pyruvate
Valine
Alanine
Leucine
ASPARTATE FAMILY
Oxaloacetate
Aspartate
Asparagine
Methionine
Threonine
Lysine
Isoleucine
AROMATIC FAMILY
Phosphoenolpyruvate
+
Erythrose 4-phosphate
Phenylalanine
Tyrosine
Tryptophan
Tyrosine
HISTIDINE FAMILY
Ribose 5-phosphate
Histidine
SERINE FAMILY
3-Phosphoglycerate
Serine
Cysteine
Glycine

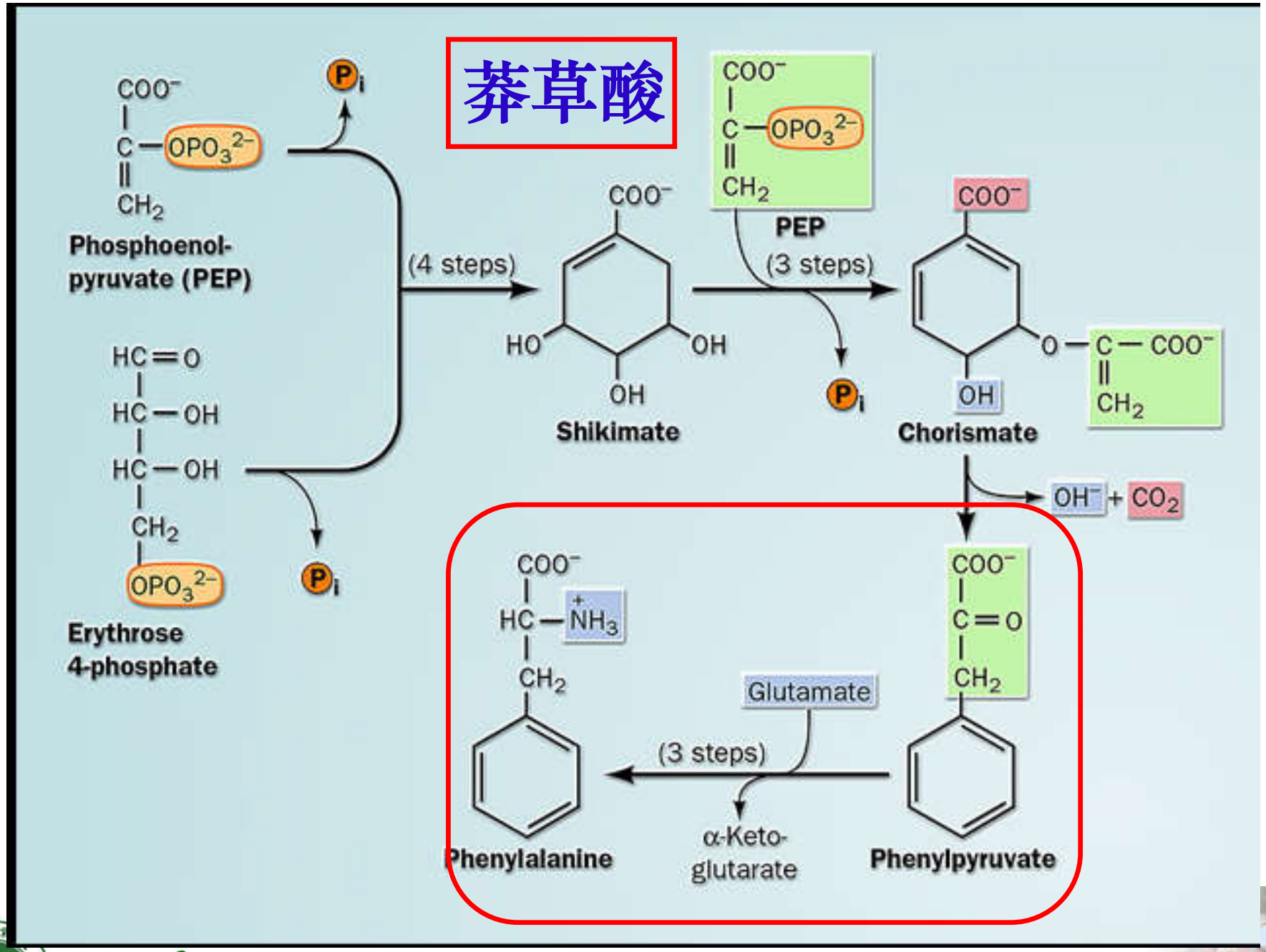

Tetrahydrofolate
THF
一碳集团（代谢）：在代谢过程中，某些化合物可以分解产生具有一个碳原子的集团，称为一碳基团，辅酶是FH4/THF （四氢叶酸）。
Serine hydroxymethyl transferase
COO⁻
HC—NH3⁺
CH2
OH
Serine
COO⁻
HC—NH3⁺
H
Glycine
N5, N10 – Methylene – THF
+
H2O

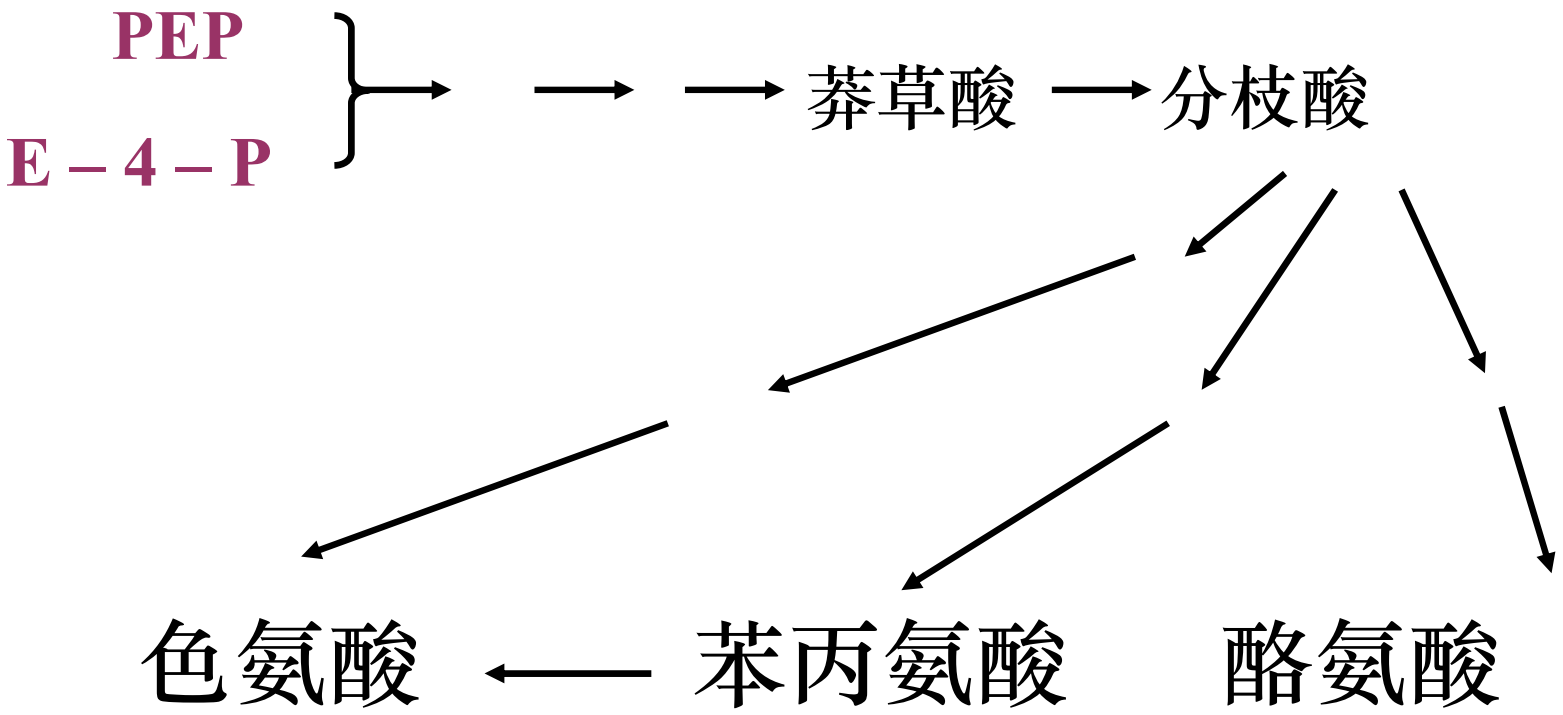

Serine acetyltransferase
Acetyl–CoA
CoASH
O-Acetylserine thiol lyase
H_2S
COO^-
NH_3^+
HC
CH_2
OH
Serine
O
C=O
CH_3
O-Acetylserine
CH_3
SH
Cysteine

Sulfur Sources for Cysteine


NH2
N
N
N
N
O
O
O
O
O
O
O
O
O
O
O
O
OH
O
O
O
O
O
O
O
O
O
O
O
O

芳香族氨基酸的合成：莽草酸途径

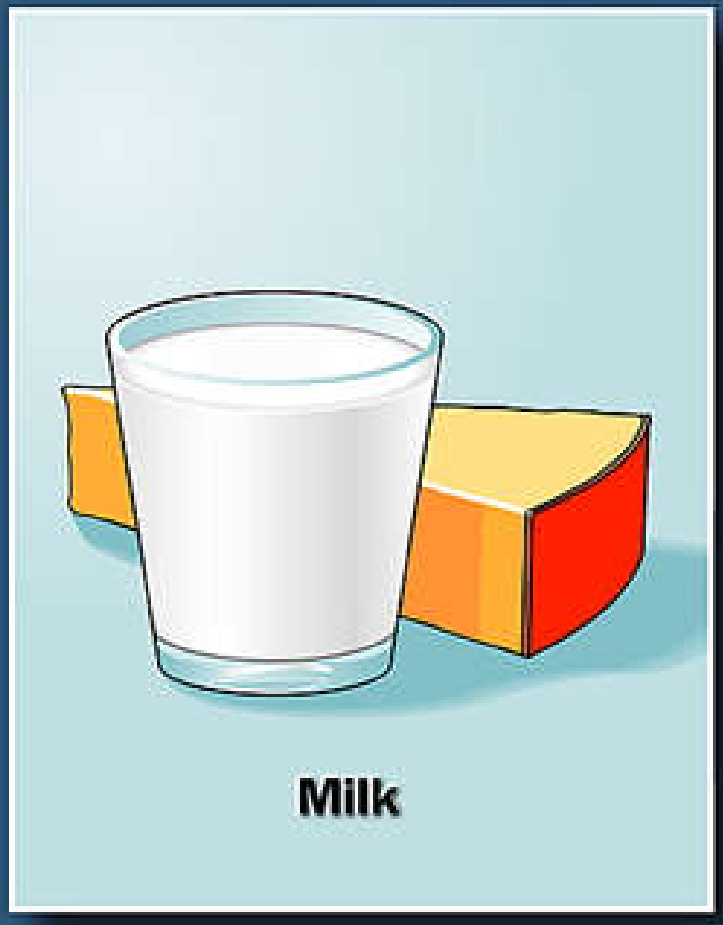


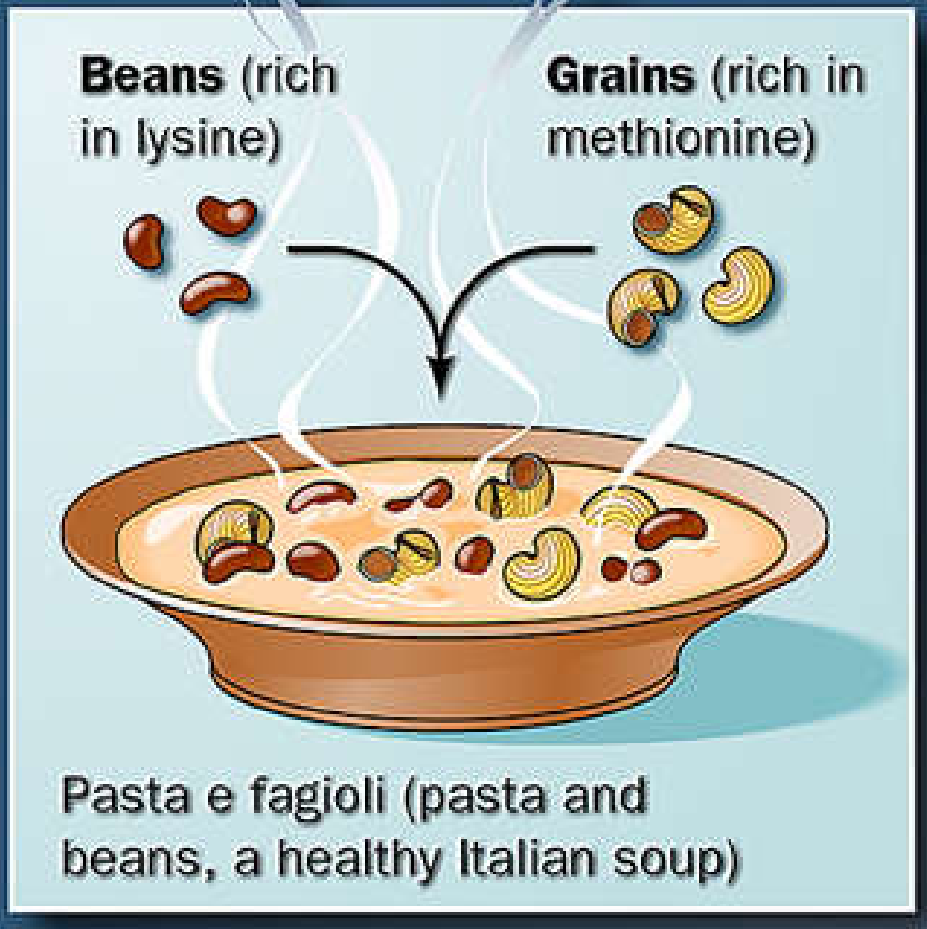

PEP
E – 4 – P
莽草酸
分枝酸
色氨酸
苯丙氨酸
酪氨酸

Only about half of the amino acids can be synthesized by us human being: the rest have to be obtained from diet, thus called **essential amino acids.**



Essential amino acids





Amino Acids that can not be synthesized by human (essential Amino acids)

9种人体必需氨基酸的简单记忆方法：
“甲组携来一本亮色书”
就是甲硫氨酸、组氨酸、
缬氨酸、赖氨酸、
异亮氨酸、苯丙氨酸、
亮氨酸、色氨酸、苏氨酸

TABLE 18-1 Nonessential and Essential Amino Acids for Humans and the Albino Rat

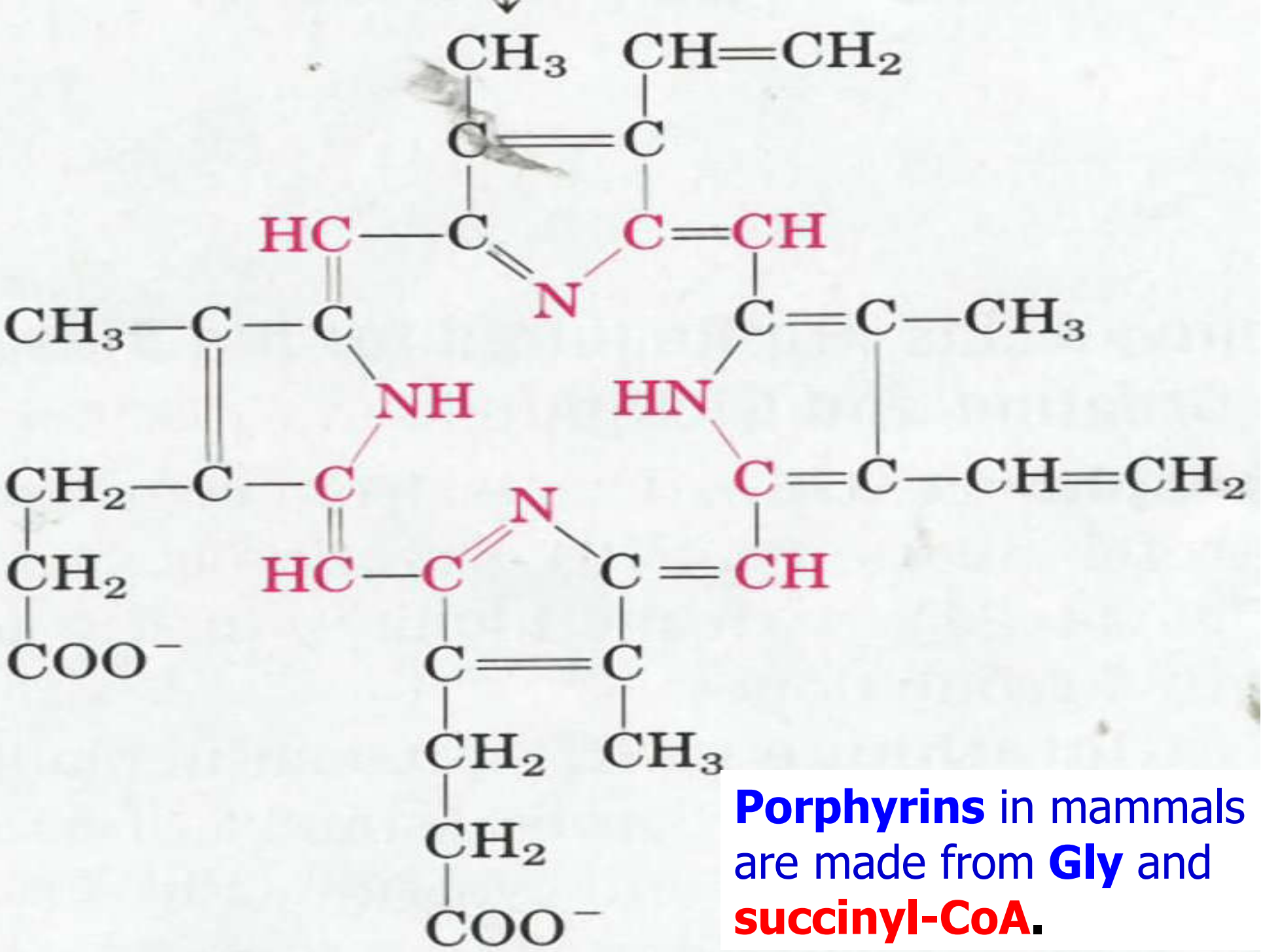
Nonessential	Conditionally essential[a]	Essential
Alanine	Arginine	Histidine
Asparagine	Cysteine	Isoleucine
Aspartate	Glutamine	Leucine
Glutamate	Glycine	Lysine
Serine	Proline	Methionine
	Tyrosine	Phenylalanine
		Threonine
		Tryptophan
		Valine

[a] Required to some degree in young, growing animals and/or sometimes during illness.



Amino acids are precursors of many specialized biomolecules





Protoporphyrin IX

Porphyrins in mammals are made from **Gly** and **succinyl-CoA**.

1. **Phosphocreatine**, derived from **creatine**. Creatine is synthesized from glycine and arginine; methionine, in the form of *SAM*.
2. Phenylalanine, tyrosine, and tryptophan are converted to a variety of important compounds in plants. **lignin**, derived from phenylalanine and tyrosine; Tryptophan is also the precursor of the plant growth hormone indole-3-acetate, or **auxin.**
3. **Nitric oxide** is synthesized from arginine in an NADPH-dependent reaction catalyzed by nitric oxide synthase in animals.



M2

Amino acid degradation



氨基酸的分解与转化

1.脱氨基作用

2.脱羧基作用

3.氨基酸分解产物的转化



氨基酸的脱氨基作用

氧化脱氨基作用

转氨脱氨基作用

联合脱氨基作用

非氧化脱氨基作用

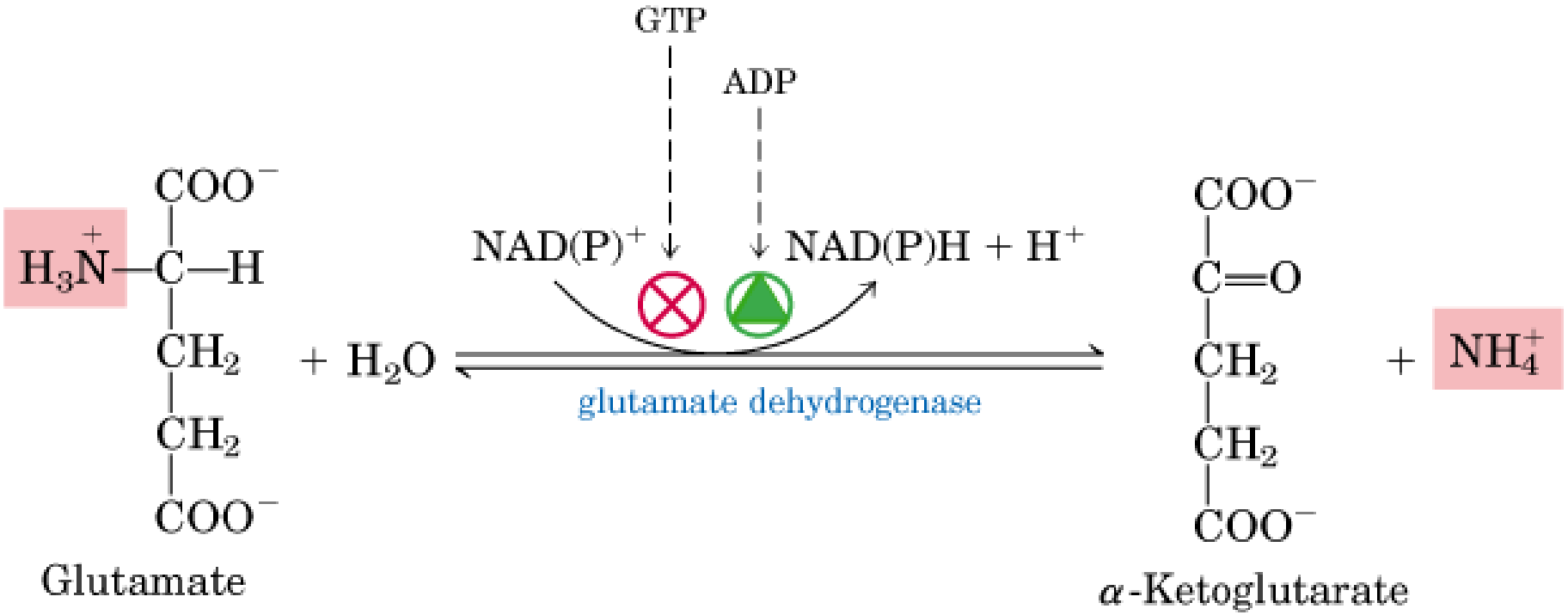
脱酰氨作用

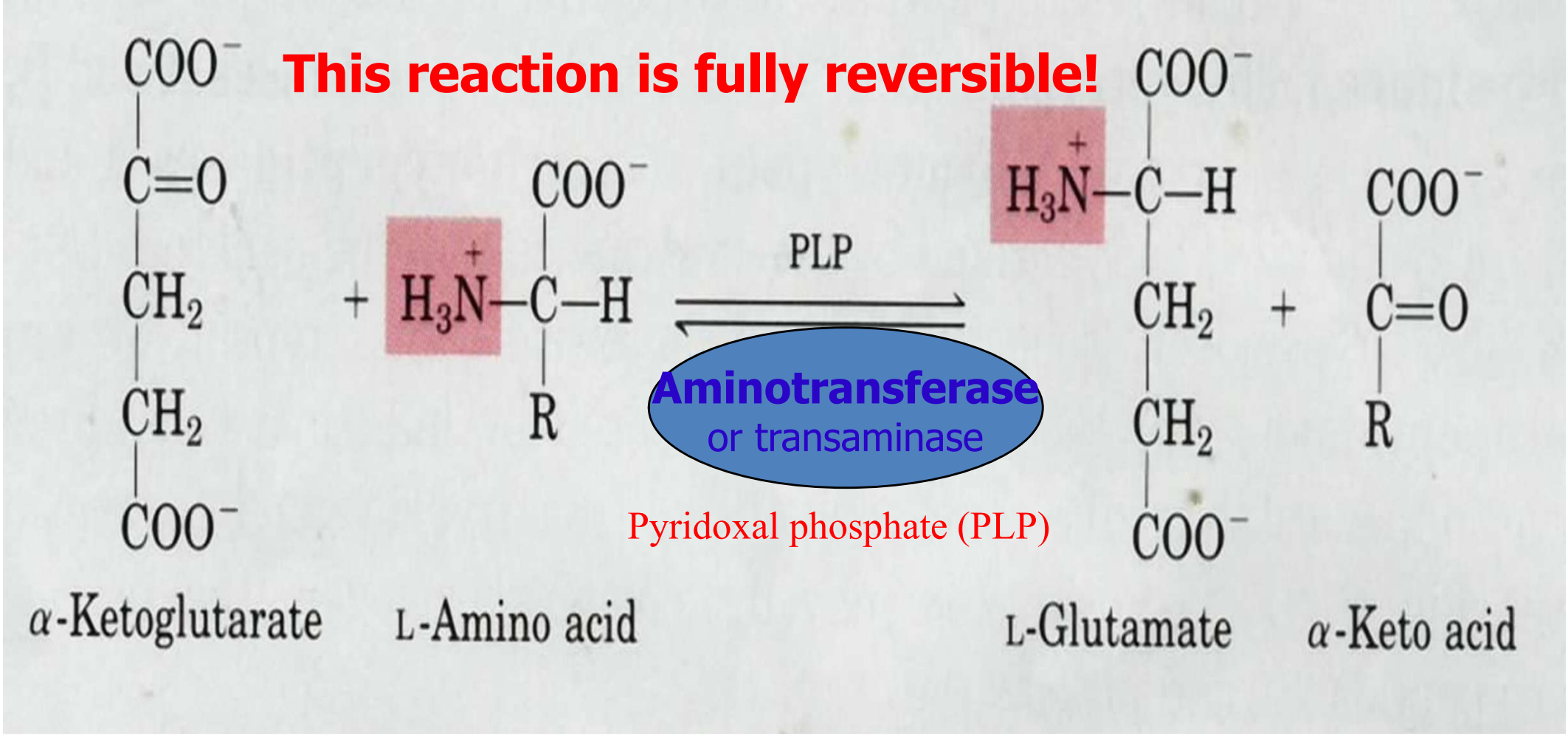


The surplus amino acids in animals can be completely oxidized or converted to other storable fuels

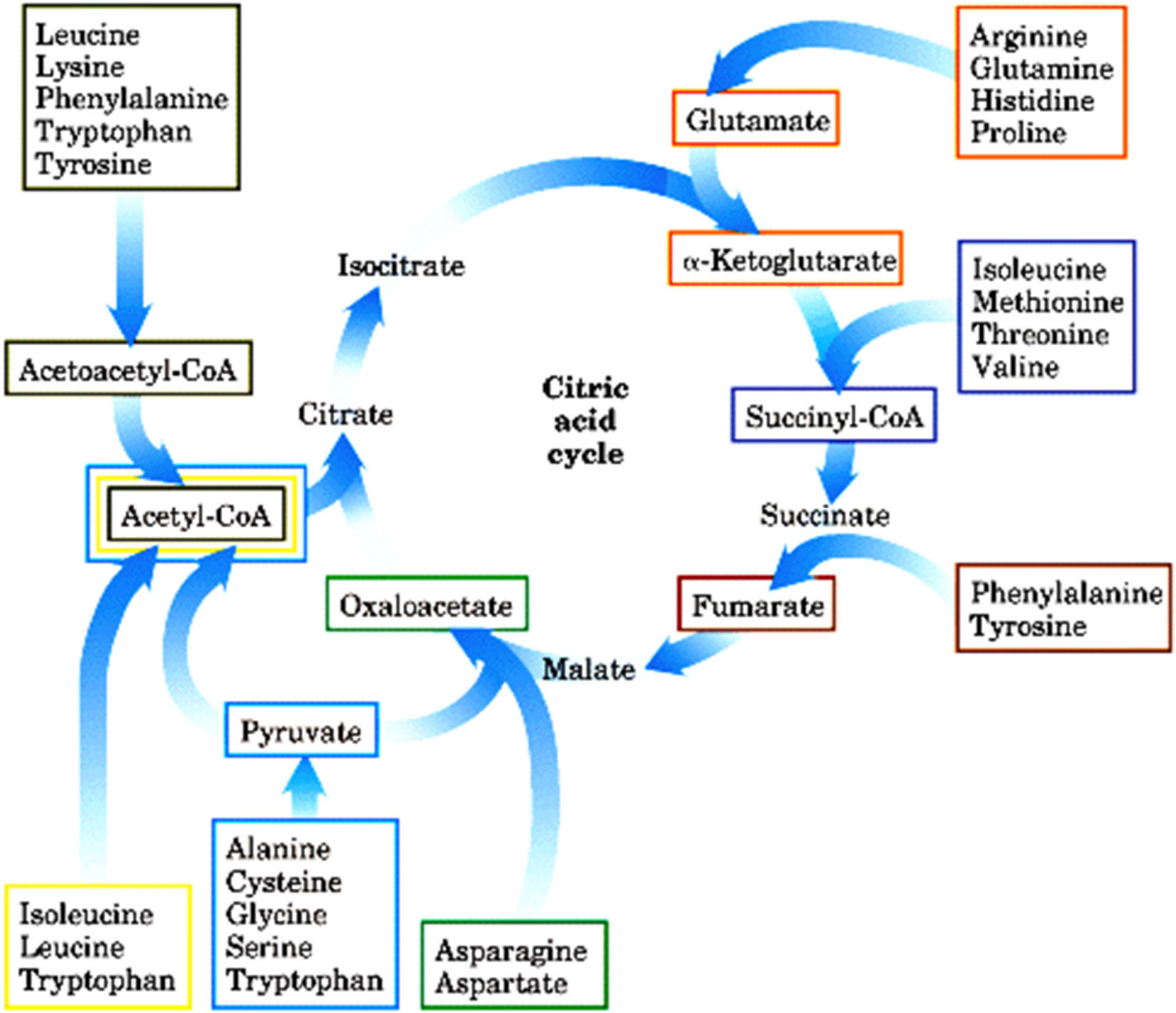
- Amino acids in excess can neither be stored, nor excreted, but oxidized to release energy or converted to fatty acids/glucose.
- Animals utilize amino acids for energy generation following a protein meal, during starvation.
- Microorganisms can also use amino acids as an energy source when the supply is in excess.
- **Plants almost never use amino acids (neither fatty acids) as an energy source.**（种子萌发例外，此时使用脂肪、蛋白质的能量）




GTP
ADP
NAD(P)+
NAD(P)H + H+
+ H2O
glutamate dehydrogenase
+ NH4+
Glutamate
α-Ketoglutarate



The α-amino group of many amino acids is transferred to **α-ketoglutarate** via catalysis of a specific **aminotransferase**, producing **Glu** and an α-keto acid.


Leucine
Lysine
Phenylalanine
Tryptophan
Tyrosine
Arginine
Glutamine
Histidine
Proline
Glutamate
Isocitrate
α-Ketoglutarate
Isoleucine
Methionine
Threonine
Valine
Acetoacetyl-CoA
Citrate
Citric acid cycle
Succinyl-CoA
Acetyl-CoA
Succinate
Oxaloacetate
Fumarate
Phenylalanine
Tyrosine
Malate
Pyruvate
Alanine
Cysteine
Glycine
Serine
Tryptophan
Isoleucine
Leucine
Tryptophan
Asparagine
Aspartate

The major route for the deamination of amino acids is transamination followed by the oxidative deamination of glutamate.（联合脱氨基）

However, a minor route also exists that involves direct oxidation of the amino acid by L-amino acid oxidase.



脱氨基产物α-酮酸的代谢转变

(1) 再氨基化生成氨基酸

(2) 转变成糖或脂肪

生糖氨基酸和生酮氨基酸

(3) 氧化成CO_2和H_2O



Trp, Ala, Gly, Ser, Cys are converted to pyruvate
(thus being glucogenic生糖氨基酸)

Part of Trp, Phe, Tyr, Leu & Lys are converted to acetoacetyl-CoA, and Part of Trp, Leu, and Ile is converted to acetyl-CoA.
(thus being ketogenic生酮氨基酸)

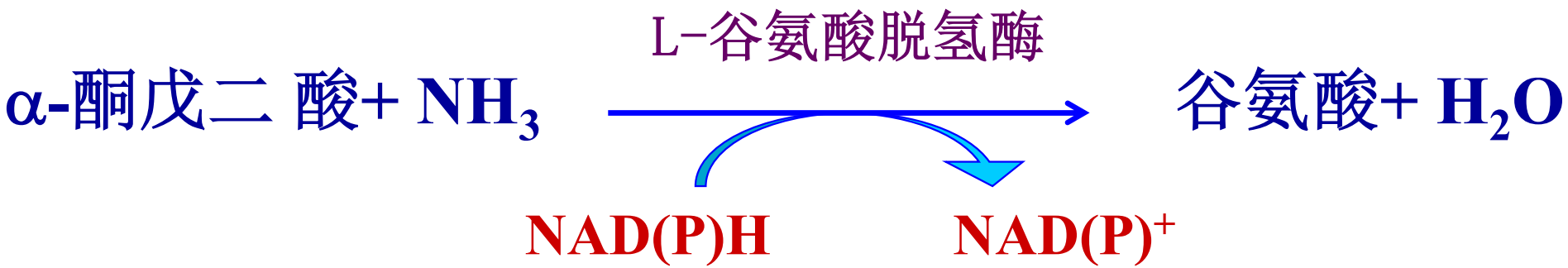


氨的代谢转变

- 重新生成氨基酸
- 谷氨酰胺和天冬酰氨的生成
- 合成其他含N物质
- 尿素的生成、及直接排氨



a、谷氨酸的重新生成



b、谷氨酰胺和天冬酰氨的生成



c、生成铵盐

与柠檬酸、苹果酸、草酰乙酸等生成铵盐，保持细胞内正常的pH

D The Urea Cycle

Excess nitrogen is excreted as ammonia. Ammonotelic (排氨的)organisms excrete **ammonia** directly, uricotelic (排尿酸的) organisms excrete it as **uric acid**, and ureotelic(排尿的) organism excrete it as **urea**.



尿素的生成

a、概念

在排尿动物体内由NH_3合成尿素是在肝脏中通过鸟氨酸循环机制完成的，这一个循环称为尿素循环。

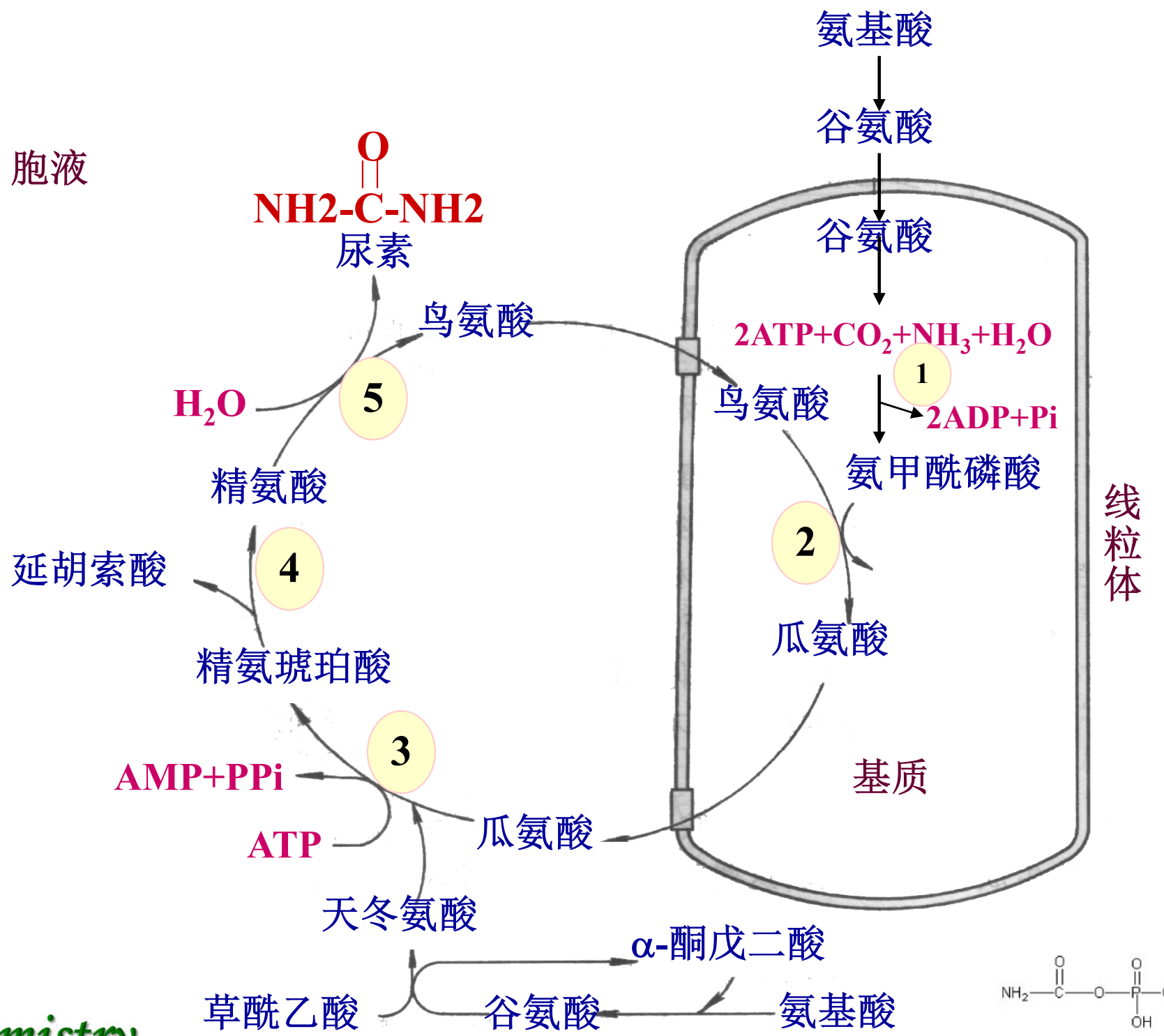
b、总反应和过程

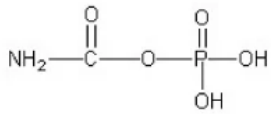
$NH_3+CO_2+3ATP+$天冬氨酸$+2H_2O \rightarrow NH_2\text{-}CO\text{-}NH_2$

$+2ADP+2+AMP+PPi+$延胡索酸



鸟氨酸循环





氨甲酰磷酸

Biochemistry

2.氨基酸的脱羧基作用

- **概念**

  氨基酸在脱羧酶的作用下脱掉羧基生成相应的一级胺类化合物的作用。脱羧酶的辅酶为磷酸吡哆醛(Pyridoxal phosphate;PLP)。

- **类型:**
  - **直接脱羧——胺**
  - **羟化脱羧——羟胺**

- **脱羧产物的进一步转化（次生物质代谢）**



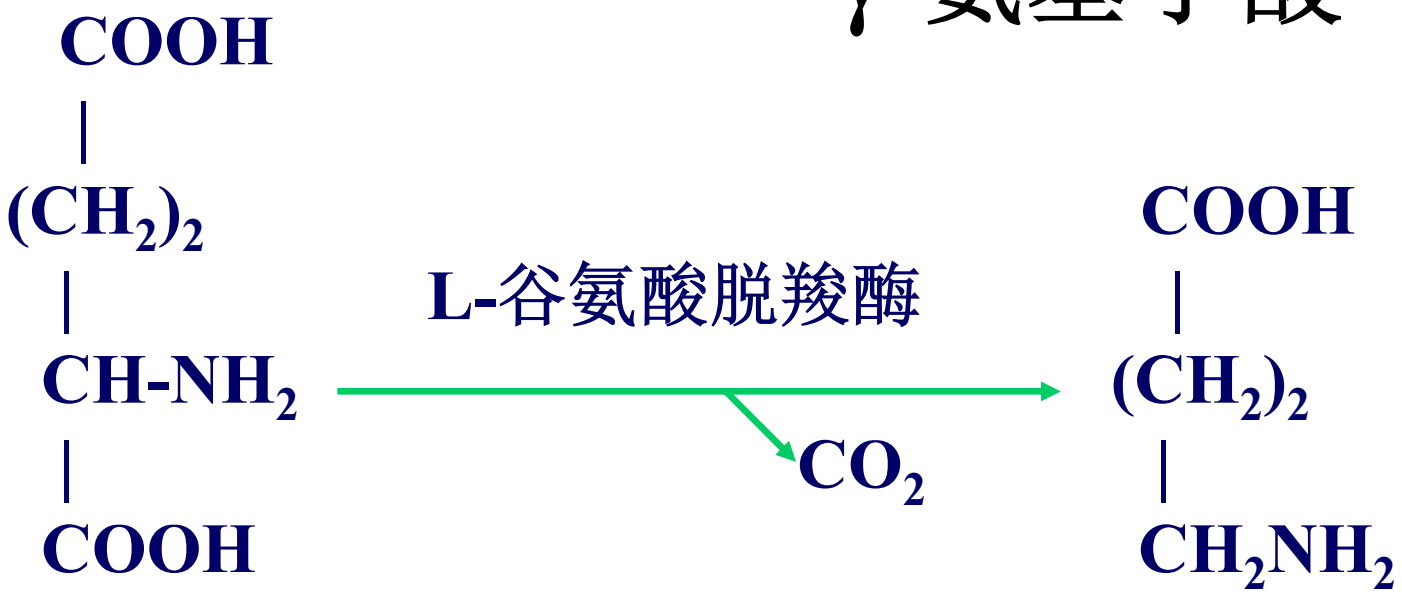
（1）直接脱羧作用

脱羧酶（辅酶为磷酸吡哆醛/胺）

- Glu→氨基丁酸（GABA）
- His→组胺
- Trp→5羟色胺
- Cys→牛磺酸
- Orn→多胺(腐胺、精脒、精胺)

腐败细菌作用时，常发生此类脱羧反应，生成胺类

γ-氨基丁酸

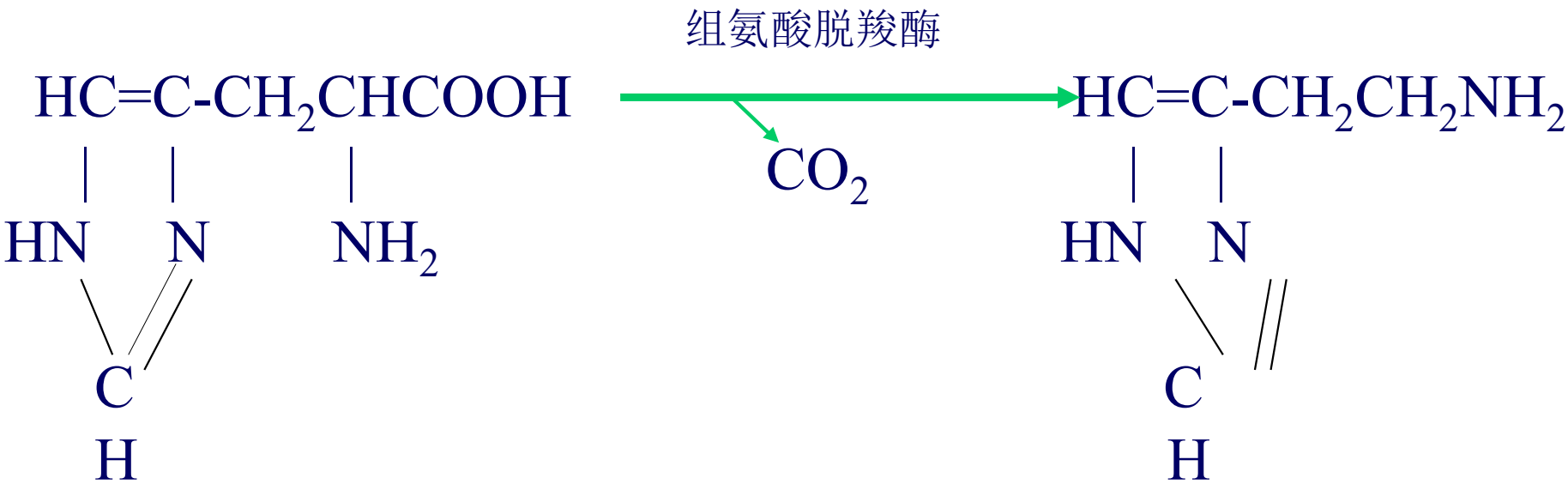


L- 谷氨酸　　γ-氨基丁酸

γ-氨基丁酸:γ-aminobutyric acid, GABA

抑制性神经递质

组胺



L-组氨酸　　组胺

组胺：血管舒张剂、胃酸刺激剂



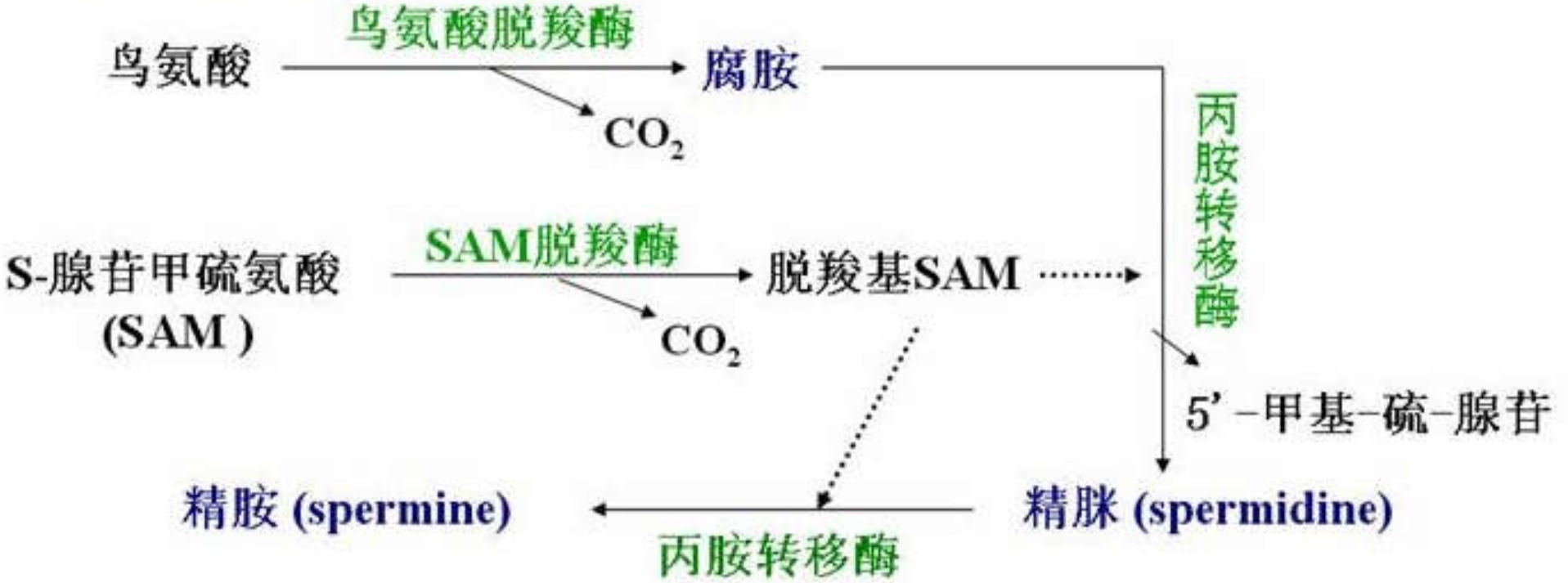
5-羟色胺 (5-hydroxytryptamine, 5-HT)

色氨酸 $\xrightarrow{\text{色氨酸羟化酶}}$ 5-羟色氨酸 $\xrightarrow{\text{5-羟色氨酸脱羧酶}}$ 5-HT + CO_2

- 5-HT在脑内作为神经递质，起抑制作用；在外周组织有收缩血管的作用。

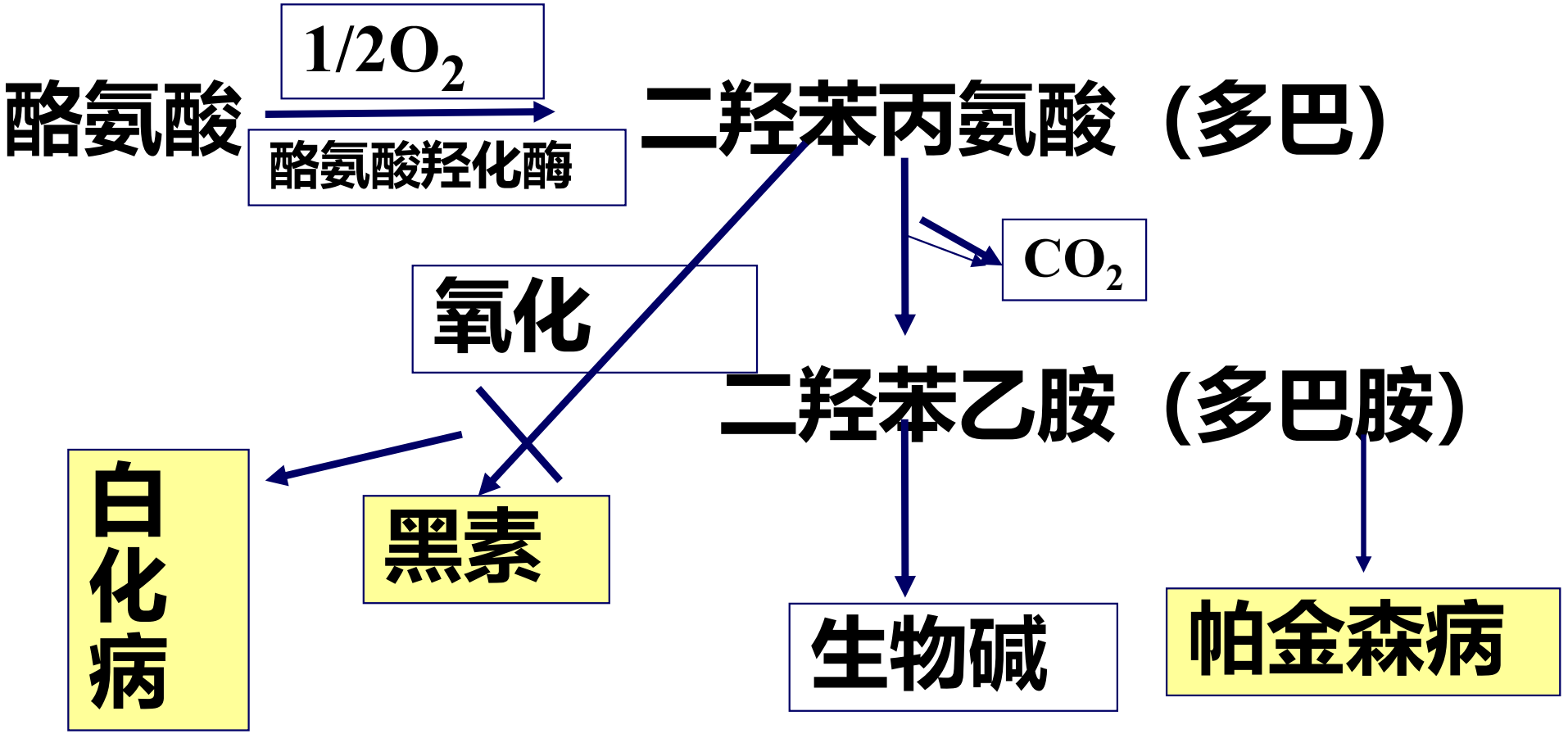


多胺(polyamines)



- 多胺是调节细胞生长的重要物质。在生长旺盛的组织（如胚胎、再生肝、肿瘤组织）含量较高，其限速酶鸟氨酸脱羧酶活性较强。**促进细胞增殖**

羟化脱羧基作用



Summary

- Nitrogen fixation： Atmospheric N_2 is reduced to ammonia by the dinitrogenase reductase and the dinitrogenase (containing a key Fe-Mo cofactor) of the **nitrogenase complex** present only in certain bacteria.
- Ammonia assimilation： Ammonia enters organic molecules via **Glu (glutamate dehydrogenase, glutamate synthase)** and **Gln (glutamine synthetase)**.
- **Glutamine/Glutamate aminotransferases** catalyzes the transferring of the amino group to many acceptor molecules.



For Biosynthesis

- Amino acids are mainly derived from intermediates of glycolysis, the citric acid cycle, and the pentose phosphate pathway.
- Pro and Arg are derived from Glu, which is synthesized from α-ketoglutarate.
- Ser, Gly, and Cys are derived from 3-phosphoglycerate.
- Met and Thr are derived from oxaloacetate.
- Lys and Ile are derived from oxaloacetate and pyruvate.



- Val and Leu are derived from pyruvate.
- Ile and Val are derived from Thr/pyruvate and two molecules of pyruvate respectively, using the same enzymes; Leu is derived from two molecules of pyruvate, sharing four steps of reactions with Val synthesis.
- Tryptophan is derived from phosphoenolpyruvate, erythrose 4-P, Gln, PRPP, and one Ser.
- Phe and Tyr are synthesized from two phosphoenolpyruvates, one erythrose 4-P （ shikimate 莽草酸）, and one Glu.



For Degradation

- Amino acid in excess can neither be stored, nor excreted, but oxidized or converted.
- The amino groups and carbon skeletons of amino acids take separate but interconnected pathways.
- Glutamate collects and delivers free ammonia（转氨基作用）.
- Gln and Glu releases NH_4^+ in mitochondria（氧化脱氨基作用）.



END

Thank you

